

Disproof of TLMC Conjecture 00000000556: Trivial HH^1 versus representation-finite rigid algebras

Abstract

Conjecture 00000000556 states that $HH^1(\Lambda) = 0$ if and only if Λ is representation-finite and rigid. We refute the “if” direction with the dual numbers $\Lambda = k[x]/(x^2)$: they are representation-finite and rigid (self-injective), yet $HH^1(\Lambda) \cong k \neq 0$.

1 The conjecture

$HH^1(\Lambda) = 0$ if and only if Λ is representation-finite and rigid.

Here $HH^1(\Lambda) = \text{Der}_k(\Lambda)/\text{Inn}(\Lambda)$ is the first Hochschild cohomology group: derivations modulo inner derivations.

2 The counterexample

Let $\Lambda = k[x]/(x^2)$ be the algebra of dual numbers over an arbitrary field k , with basis $1, x$ and $x^2 = 0$.

Attack 1 (Representation-finite).

A Λ -module is a pair (V, T) where T is a nilpotent endomorphism of the vector space V with $T^2 = 0$. Over an algebraically closed field such pairs decompose into Jordan blocks of size at most 2, hence the only indecomposable Λ -modules are k and Λ itself. There are exactly two, so Λ is representation-finite.

Attack 2 (Rigid, i.e. self-injective).

The linear form $\lambda(a + bx) = b$ has associated bilinear form $B(u, v) = \lambda(uv)$, whose matrix on the basis $\{1, x\}$ is $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ with determinant -1 (1 in characteristic 2) — never zero. Hence Λ is a Frobenius algebra, therefore self-injective, i.e. rigid. (Independently, $\text{Ext}_{\Lambda}^1(\Lambda, \Lambda) = 0$ trivially, since Λ is projective as a module over itself.)

Attack 3 ($HH^1(\Lambda) \neq 0$).

Let $d: \Lambda \rightarrow \Lambda$ be k -linear with $d(1) = 0$ and $d(x) = v$. The Leibniz rule applied to $x^2 = 0$ forces $2xv = 0$; conversely any such v yields a derivation.

Case $\text{char } k \neq 2$: $2xv = 0$ forces $v \in (x)$, so $\text{Der}_k(\Lambda) \cong k$, spanned by $d = x\partial$, i.e. $d(1) = 0$, $d(x) = x$.

Case $\text{char } k = 2$: the constraint is vacuous, $\text{Der}_k(\Lambda) \cong k^2$.

Since Λ is commutative, $uv - vu = 0$ for all u, v , so every inner derivation vanishes: $\text{Inn}(\Lambda) = 0$. The element $d = x\partial$ satisfies $d(x) = x \neq 0$, hence is a derivation which is *not* inner. Therefore

$$HH^1(\Lambda) \cong \begin{cases} k & \text{char } k \neq 2, \\ k^2 & \text{char } k = 2, \end{cases} \quad \text{in particular } HH^1(\Lambda) \neq 0.$$

3 Conclusion

$\Lambda = k[x]/(x^2)$ is representation-finite and rigid, but $HH^1(\Lambda) \cong k \neq 0$. The direction “representation-finite and rigid $\Rightarrow HH^1 = 0$ ” of Conjecture 00000000556 fails, and with it the conjectured equivalence.

4 Boundary of the attack

- **Characteristic.** The attack works over every field: $HH^1 \cong k$ for $\text{char } k \neq 2$ and $HH^1 \cong k^2$ for $\text{char } k = 2$ — nonzero either way — and even over $\mathbb{Z}$.
- **Meaning of “rigid”.** The refutation adopts the self-injectivity reading, which the dual numbers satisfy (Frobenius). Under two other readings Λ would *not* be rigid and this particular attack would not apply: (i) deformation-rigidity ($HH^2 = 0$): the 2-periodic Λ^e -resolution with differentials $x \otimes 1 - 1 \otimes x$ and $x \otimes 1 + 1 \otimes x$ gives $HH^2(\Lambda) \cong \Lambda/(x) \cong k \neq 0$; (ii) “every indecomposable module is rigid”: $\text{Ext}_{\Lambda}^1(k, k) \cong k \neq 0$ via the nonsplit extension $0 \rightarrow (x) \rightarrow \Lambda \rightarrow k \rightarrow 0$.
- **Failure direction.** Only the “ $\Leftarrow$ ” direction is refuted; nothing is claimed about “ $\Rightarrow$ ”.

5 Verification

- `reproduce.py`: independent exact linear-algebra recomputation over $\mathbb{Q}, \mathbb{F}_2, \mathbb{F}_3$: $\dim \text{Der} = 1/2/1$, $\dim \text{Inn} = 0$, $\dim HH^1 = 1/2/1$, Frobenius determinant $\neq 0$, and $d(x) = x$ is a derivation.
- `lean4/`: formal certificate in core Lean 4.33 (no Mathlib, no dependencies). `lake build && lake env lean Check.lean` prints “does not depend on any axioms” for every theorem: zero axioms, zero `sorry`. It formalizes that $x\partial$ is a derivation, that all inner derivations vanish, that $HH^1 \neq 0$, the classification $f(a, b) = (0, b \cdot f(x)_2)$ (so $\text{Der} \cong \text{base ring}$), and the nondegeneracy of the Frobenius form.